

Dreaming

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Differences in Emotional Content, Dream Awareness, and Dream Control Between Lucid Dreams and Out-of-Body Experiences: Report Analyses

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Lucid dreams (LDs) and out-of-body experiences (OBEs) initiated from sleep paralysis both involve heightened cognitive abilities during sleep. Some researchers suggest that OBEs, specifically originating from sleep paralysis, are closely related

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Francisco Tomás Gallo served as lead for research, data acquisition, manual scoring, statistical analysis, and writing—original draft and contributed equally to the conception and conceptualization of the study. Cecilia Forcato contributed equally to the conception and conceptualization of the work, critically revising the article, and served as lead for project administration and funding acquisition. Nerea Lucía Herrero supported in data acquisition. Daniela Godoy and Antonela Tommasel performed text processing and the natural language processing analysis. Ignacio Spiousas contributed equally to the statistical analysis. Miguel Gasca-Rolin, Rodrigo Ramele, and Pablo M. Gleiser contributed equally to the critical revision of the article. Rodrigo Ramele also supported in funding acquisition.

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to LDs or even considered variations of the same phenomenon. Our study compared non-LDs, LDs, and sleep paralysis-induced OBEs from 60 participants over 2 months (916 dream reports). We used Lexicons like National Research Council Canada Emotion Lexicon and Empath, along with Hall and Van de Castle's scoring system (Domhoff, 1996) with variations and additional measures. The results showed that OBE were characterized by higher occurrences of negative emotions compared to both lucid and non-LDs as measured by automatic and manual scoring systems. Also, more OBE reports contained words related to agency and insight; higher manual scoring of dream control-related expressions; and more total sensations, dream activities, reference to prospective memory evocations, spatial and body references, and more difficulties with movement within the dream environment, than LDs. The findings support the idea that sleep paralysis-induced OBEs represent unique experiences distinguished from LDs.

Keywords: lucid dreams, out-of-body experiences, emotional content, dream awareness, dream control

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Dreams, an intricate facet of the sleeping mind (Siclari et al., 2013), present an intricate challenge for scientific research due to their inherently subjective nature. To unravel the enigma of dreaming, researchers must largely rely on postawakening retrospective reports (Rosen, 2013). Within the realm of conscious dream experiences, two distinct types have garnered significant attention: lucid dreams (LDs) and sleep paralysis-induced out-of-body experiences (OBEs). Both LDs and OBEs involve the reemergence of higher cognitive functions during sleep, including reflective self-awareness and abstract thinking (Dresler et al., 2014; Herrero et al., 2023). Notably, LDs entail dreamers recognizing their dream state and often exercising control over the dream narrative (Holzinger & Mayer, 2020; S. P. LaBerge, 1980). Typically, these experiences manifest during rapid eye movement sleep, accompanied by heightened low gamma oscillatory activity (40 Hz) in the frontal and temporal regions of the brain (S. Mota-Rolim et al., 2008; Voss et al., 2009; although see Baird et al., 2022). The OBEs, on the other hand, involve the sensation of being outside of one's physical body and viewing the world from an external perspective (Blackmore, 1982; Blanke & Arzy, 2005; Cheyne, 2003; de Sá & Mota-Rolim, 2016; Herrero et al., 2023; Irwin, 1988; S. P. LaBerge, 1988; Sheils, 1978). OBEs are often reported as extremely vivid and can have the qualities of veridical perception (Blackmore, 1982; Herrero et al., 2023; S. A. Mota-Rolim et al., 2020). The sleep stage as well as the predominant EEG oscillations involved in OBEs are still unknown. However, it is thought to occur during sleep paralysis (SP), which is characterized by an inability to move voluntarily and a feeling of awareness of one's surroundings (Cheyne, 2003; Hishikawa & Shimizu, 1995). SP occurs during the transition between rapid eye movement sleep and wakefulness (Terzaghi et al., 2012), and it is associated with mixed alpha and theta brain waves (Mainieri et al., 2021; Takeuchi et al., 1992, 2002). Thus, although OBEs are thought to occur during SP, the categorization of OBEs is still a topic of debate among scientists, with some classifying them as a type of LD (S. P. LaBerge, 1985; S. LaBerge & DeGracia, 2000;

Levitán et al., 1999), while others view them as a related phenomenon to LD (Holzinger & Mayer, 2020; Raduga et al., 2020; Yu & Shen, 2020). Some authors even do not make any distinction between LD and OBE reports (Dodet et al., 2015). Despite their potential implications, the infrequent occurrence of these experiences, particularly OBEs, presents significant challenges when attempting to study them within a laboratory setting.

Dream content analysis is a crucial aspect of sleep research, offering valuable insights into the subjective experiences during sleep (see Elce et al., 2021 for a complete review). Traditional methods involve collecting free reports or administering questionnaires, each with its strengths and limitations. Free reports, obtained through dream diaries, provide a comprehensive view but face challenges like intersubject differences and subjective extraction of quantitative data. Analyzing dream content typically involves extracting structural and semantic information, relying on manual scoring by human raters. However, recent advancements in computational linguistic analysis present an opportunity to overcome these limitations (Altszyler et al., 2017). By leveraging linguistic features, these methods offer a more objective and reproducible assessment (Schwartz, 2004).

Building upon these advancements in dream content analysis, we aim to explore the variations in dream content among individuals who have never experienced neither LDs nor OBEs (nonlucid dreamers), those who have experienced LDs but not OBEs (lucid dreamers), and those who have experienced OBEs (OBE dreamers) with or without LDs. In total, 916 dream reports (731 non-LDs, 117 LDs, and 68 OBE) of 60 individuals were analyzed using automatic and manual analysis methods, providing a comprehensive examination of the experiences reported by each group.

Method

Participants

Our recruitment strategy utilized both social media outreach and the distribution of flyers within relevant Facebook groups. These combined efforts successfully connected us with potential participants, ultimately resulting in 239 interviews conducted. These interviews were carried out virtually via Google Meet, during which we extensively inquired about participants' sleep patterns and dream experiences. After the interviews, participants were provided with an instruction manual and a convenient Google Drive link, facilitating the easy upload of their dream records. From the pool of 143 individuals who expressed interest in the study and completed this initial step, only 60 participants remained committed and successfully maintained their dream diary over the 2-month study duration.

The participants, who were all native Spanish speakers from different countries, almost all from Argentina (there were single participants from Colombia, Venezuela, Spain, and Mexico), were categorized into three distinct groups: nonlucid dreamers (never experienced an LD nor an OBE), lucid dreamers (lucid dreamer who never experienced an OBE), and OBE dreamers (has experienced OBEs during SP at least once in their lifetimes, with or without LDs). The nonlucids had an age of 31.8 ± 10.6 (14 subjects, six male and eight female), the lucids 33.2 ± 8.5 (24 participants, eight male and 16 female), and the OBE 35.1 ± 11.2 (22 participants, 10 male, 11 female, and one nonbinary). We obtained informed consent from all subjects prior

to their participation in the study and provided all participants with a written explanation of the study procedures and their right to withdraw at any time without penalty. This study was approved by the Biomedical Research Ethics Committee of Alberto C. Taquini Institute for Translational Medicine Research, in accordance with the principles expressed in the Declaration of Helsinki.

Dream Journal

Participants were asked to complete a dream journal for 2 months. They were instructed to write down every dream they experienced right after the morning awakening, regardless of whether it was lucid or not. Participants were encouraged to record all details they could recall, ranging from the plot and characters to the emotions and sensations experienced during the dream. They were also instructed to include the time and date, their level of awareness during the dream, and how they gained lucidity (if applicable).

Dream Classification

Two independent taggers classified dreams based on descriptions provided in a journal. A dream report was considered lucid if the dreamer was either directly or indirectly aware. An OBE was identified if the dreamer described leaving the body or reported an aura (described in Herrero et al., 2023). We discarded 14 vague or unspecific reports, resulting in a sample of 916 dreams (731 non-LDs, 117 LDs, and 68 OBE).

Text Processing and Analysis for Dream Reports

The text of dream reports was processed using Natural Language Processing techniques and using the spaCy library (<https://spacy.io/>) with pretrained models for Spanish (the “es_core_news_sm” model, a small model trained with news text). Tokenization was initially conducted to break down the text into meaningful elements or tokens (i.e., words). Subsequently, part-of-speech (POS) tagging was applied to assign specific POSs (such as noun, verb, adjective, adverb, etc.) to each word. This step is pivotal in Natural Language Processing as it furnishes fundamental insights into the language’s structure and semantics. Notably, for our analysis, only nouns, verbs, adjectives, and adverbs were retained, as these POSs are demonstrated to harbor the most pertinent information for various text-related tasks, as shown by Bonatti et al. (2016). For further information on dream analysis, interested readers may refer to Altszyler et al. (2017), Elce et al. (2021), and Schwartz (2004).

For sentiment and content analysis, the National Research Council Canada (NRC) Emotion Lexicon (Mohammad & Turney, 2010) and Empath (Fast et al., 2016) lexicons were used. NRC Emotion Lexicon provides intensity scores for words in eight basic emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust) and two sentiments (positive and negative), while Empath covers 200 emotional and topical categories validated by humans. As the Empath lexicon is only available in English and considering the experiences reported in Perczek et al. (2000), who analyzed the reliability of Spanish translation from English for psychometric scales and lexicons, we used a translated version of Empath to Spanish. This Spanish Empath was originally generated by Tommasel et al. (2022). The authors

automatically translated the Empath categories and their associated words to Spanish. For this translation, the authors relied on the IBM Watson Language Translator. Then, the authors independently checked the resulting translations for inconsistencies. Given the gendered nature of Spanish, if a word in a category was associated with a specific gender, the authors also added the words corresponding to the other gender and a gender-neutral version (if it existed in Spanish). To allow for a comprehensive analysis of dreams, a unique value for each emotion and topical category was computed for each dream report by averaging the individual word scores. This made it possible to analyze the proportion of words referring to each emotion or category and the intensity scores of words in each category, enabling a thorough analysis of the emotional and conceptual content of the dream reports.

We also calculated the unique word count (WC) and total WC. WC is a crucial aspect to consider when examining text (Rosenlicht et al., 1994; Röver & Schredl, 2017; Schredl, 1999, 2011; Urbina, 1981).

The initial methodology of this study involved a detailed preanalysis of the text bodies generated by three distinct groups of participants: nonlucid dreamers, lucid dreamers, and OBE dreamers. This preanalysis aimed to provide an overview of the linguistic characteristics present in the texts before conducting a more in-depth analysis. The results obtained through various language analysis metrics are summarized in Table 1.

The evaluation of the type-token ratio (Baayen, 2001), a measure of lexical diversity, reveals comparable values among the groups, suggesting relative uniformity in vocabulary variability. Similarly, the examination of lexical density (Halliday & Hasan, 2014) shows similar proportions of meaningful words relative to the total words, indicating coherence in semantic expression between the three groups. Vocabulary size (Nation, 1990), in particular per document, exhibits similarities among the three groups, suggesting that lexical richness does not vary significantly between dreamers. Finally, when examining additional metrics such as entropy and average word length, no notable differences are observed among the groups. Shannon’s entropy, measuring uncertainty in the word distribution (Shannon, 1951), and average word length (Wimmer & Hummer, 1990), indicating the average size of linguistic units, consistently show similar results across participants in different groups. This finding reinforces the notion that, beyond lexical diversity, other linguistic characteristics also remain comparable among the analyzed groups.

Table 1
Language Analysis Metrics for Dreamer Groups

Metric	Nonlucid dreamers	Lucid dreamers	OBE dreamers
Type-token ratio	0.3202	0.2294	0.2766
Total words	36,272	73,725	44,310
Content words	16,228	33,745	20,769
Vocabulary size	5,320	8,477	6,104
Vocab size per document	26.3	24.5	27.6
Shannon’s entropy	11.15	11.47	11.13
Average word length	6.06	6.08	6.10
Lexical density	0.4474	0.4580	0.4687
Dreams per participant	14.6 ± 14.6	15.3 ± 21.5	15.2 ± 14.9

Note. OBE = out-of-body experience.

Agency and Insight

We used the category-level analysis of KH Coder 3 Software (Higuchi, 2016), creating two categories based on the idea of Mallett and Ashokkumar (2022). We created “agency” and “insight,” since they are considered proxies for lucid dreaming. To do so, the agency category was formed by the agency category from the Big Two dictionary (Pietraszkiewicz et al., 2019) and insight was formed by the insight category from the Linguistic Inquiry and Word Count 2015 (Pennebaker et al., 2015) dictionary, both categories with our own adaptations to the Spanish language.

Oneiric Content Analysis

In this phase of the study, dream content analysis was conducted using the Hall and Van de Castle dream content scoring system (Domhoff, 1996), a well-established methodology for examining dreams. This system comprises two primary components: the manifest content (literal description of the dream) and the latent content (psychological interpretations). It categorizes various aspects of dreams, including characters, actions, settings, emotions, and sexual themes, among others, facilitating the exploration of patterns and meanings within dream experiences. For the present study, we selectively retained certain categories from the Hall and Van de Castle system, specifically focusing on “emotions,” “sensations,” and “descriptions.” Additionally, we expanded the framework by introducing several new parameters tailored to our research context. These included “lucidity,” “dreamer activities,” “spatial” or “motor references,” and “memory type references.” The manual scoring in this study was performed blindly to avoid any bias. Additionally, the scoring was corroborated by two independent individuals who analyzed a percentage of the dream reports (15% of the total). Only those categories with consistent scoring across all three raters were included in the subsequent analysis.

Emotions. To identify and quantify emotional elements within the dream content, expressions that directly or indirectly conveyed specific emotions were selected. These emotions encompassed a range of feelings, including “anger,” “apprehension,” “sadness,” “confusion,” and “happiness.” Expressions were carefully scrutinized and assigned to the relevant emotional categories based on predefined criteria.

Sensations. Sensations within the dream reports were categorized into two main groups: “physical sensations” and “other.” The former encompassed visual, auditory, somatosensory, motor, pain related, vestibular sensations, and others. The latter, “other,” accounted for sensations that were challenging to categorize within the previously mentioned groups.

Descriptions and Spatial References. A diverse range of descriptions was examined, including those associated with characters, the dreamer, objects, and locations. Modifiers such as adjectives or complex clarifications were considered as descriptions. Spatial references, involving both personal and external spatial elements, were also documented. These references could involve either location (e.g., “he was sitting in the corner”) or specifying movement within a spatial dimension (e.g., “I turned to the right”).

Lucidity. The assessment of dream lucidity involved recognizing references to either the dream or waking states, specifically focusing on meta-references. We excluded references made during the narration and only considered those references that occurred within the dream itself and were explicitly evident in the story as such.

We quantified effective dream control by measuring actions such as intentional flight, ensuring that these actions were carried out voluntarily and not as mere elements of the dream narrative. Additionally, we assessed the recognition of dream strangeness whenever it manifested itself during the dream experience.

Dream Activities. The different types of activities carried out by the dreamer in the dream, such as physical and nonphysical actions, thoughts, verbalizations, and expressed intentions, were also quantified.

Memory Evocations. The presence of memory-related content within dreams was assessed across various types, including episodic, autobiographical, prospective, and semantic memories. Expressions reflecting recollections of events, identity, intentions, or general information from the dreamer's waking life were considered.

Embodiment and Movement Difficulties. References to the dreamer's own body and challenges in movement within the dream were noted. These references encompassed both physiological aspects and issues related to motor control, excluding supernatural or external influences.

The analysis counted expressions per report for each category, irrespective of the specific dream character involved. This approach enabled a comprehensive examination of dream content without delving into intricate character-specific details. We focused on tallying expressions per report related to each category, disregarding the attribution of actions to individual dream characters. Notably, there were no significant group differences in report length (in the Results section), reinforcing the decision to present results in terms of expressions per report rather than expressions per WC.

Statistical Analysis

The analysis was performed in two parts, one for "type of dream" and another for "type of dreamer." In the first part, we examined nonlucid dream (non-LD) reports of nonlucid dreamers (NN), the LD reports of lucid dreamers (LL), and the OBE reports of OBE dreamers (OO). The second part involved NNs, non-LDs reports of lucid dreamers (NL), and non-LDs reports of OBE dreamers (NO). This decision acknowledges the impact of training on frequent lucid dreamers' nonlucid reports and the unique perspectives of individuals with frequent OBEs.

For both subsets, a mixed effects model was constructed with the variable (e.g., anger and happiness) as the response variable and type of dreamer or type of dream as the predictor. We included a random effect term to account for individual variability among dreamers. Pairwise comparisons were subsequently performed using the emmeans function of the emmean R package (Lenth, 2024; Searle et al., 1980), facilitating a detailed examination of differences in anger levels between specific types of dreamers within each dream category. To assess the overall significance of the fixed effects, analysis of variance was conducted using the anova function. The parameters of the models and the results of the *F* test and pairwise comparisons are presented in Tables 1–3 in the online supplemental materials, including the parameters of the models, *p* values for the pairwise comparisons (corrected for multiple comparisons using the Tukey method), and effect sizes for each comparison.

In order to compare the LDs and OBEs, a mixed effects model was constructed with the variable (e.g., emotion and sensation) as the response variable and type of dream as the predictor. We also included a random effect term to account for

individual variability among dreamers. The resulting parameters and F test for the model are displayed in Tables 1–3 in the online supplemental materials.

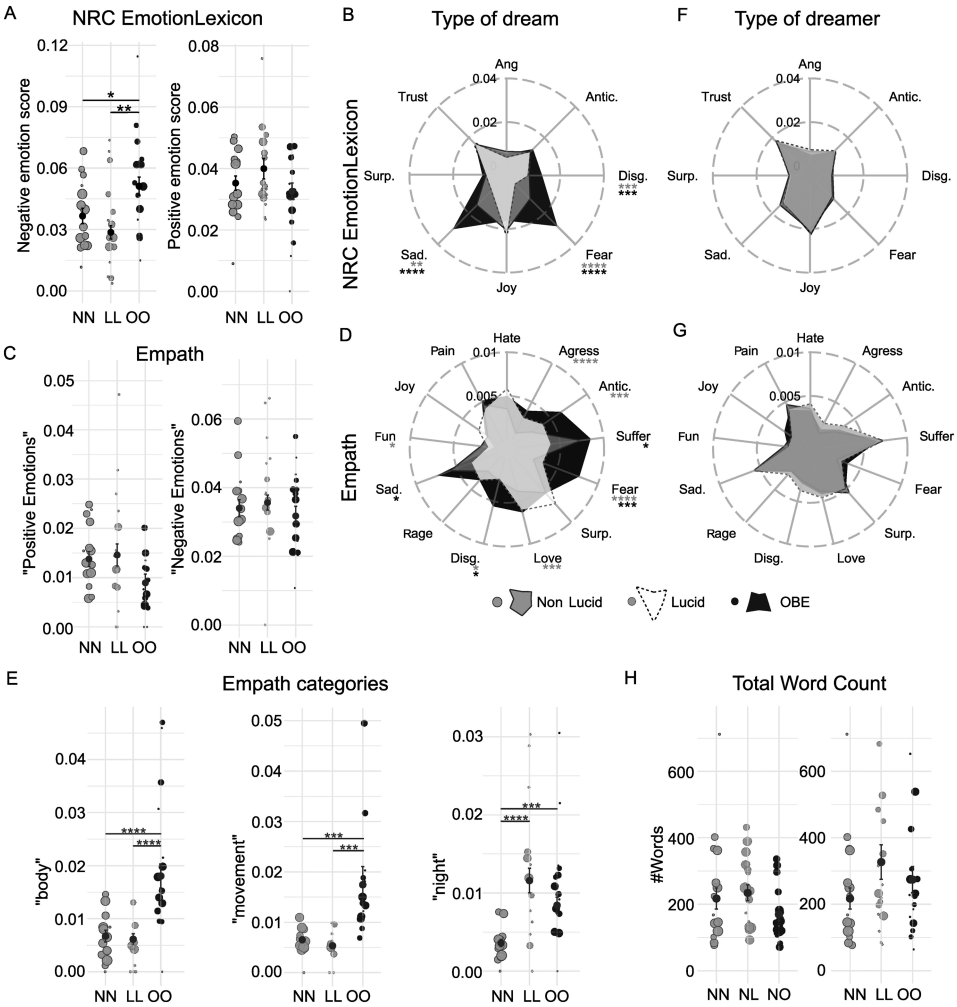
Results

We first analyzed the typical dream reports of each dreamer, namely the non-LD NN, the LL, and the OO (Figure 1). The NRC Emotion Lexicon analysis showed that the OO reports had higher negative emotions scores than the NN and LL reports, but no significant differences were found for negative score between NN and LL reports (Figure 1A), $F(2, 44) = 6.69$, $p = .003$; multiple comparisons, $p_{OOvsNN} = .031$, $p_{OOvsLL} = .003$, $p_{NNvsLL} = .389$. On the contrary, there were no significant differences for the positive emotions score (Figure 1A), $F(2, 44) = 1.18$, $p = .317$. We further analyzed with NRC Emotion Lexicon the scores of specific emotions. The scores for sadness and disgust were significantly higher in OO reports than in NN and LL reports, with no differences between the last two (Figure 1B), sadness, $F(2, 44) = 15.04$, $p < .0001$; multiple comparisons, $p_{OOvsNN} = .0025$, $p_{OOvsLL} < .0001$, $p_{NNvsLL} = .121$; disgust, $F(2, 44) = 10.64$, $p = .0003$; multiple comparisons, $p_{OOvsNN} = .0008$, $p_{OOvsLL} = .0006$, $p_{NNvsLL} = .919$. Fear emotion score was significantly higher in OO reports compared to both NN and LL reports, but no difference was found between NN and LL reports, $F(2, 44) = 25.13$, $p < .0001$; multiple comparisons, $p_{OOvsNN} = .0001$, $p_{OOvsLL} < .0001$, $p_{NNvsLL} = .058$.

The analyses of emotions using the Empath tool showed no significant difference in the negative and positive emotion scores (Figure 1C), negative, $F(2, 44) = 0.39$, $p = .681$; positive, $F(2, 44) = 2.95$, $p = .065$. However, when we analyze the specific type of emotions, higher expressions of fear and disgust were observed for OO reports compared to the other groups, but no differences were observed between NN and LL reports (Figure 1D), fear, $F(2, 44) = 15.62$, $p < .0001$; multiple comparisons, $p_{NNvsOO} < .0001$, $p_{LLvsOO} = .0018$, $p_{NNvsLL} = .737$; disgust, $F(2, 44) = 5.00$, $p = .011$; multiple comparisons, $p_{NNvsOO} = .021$, $p_{LLvsOO} = .038$, $p_{NNvsLL} = .952$. Suffering score was lower in LL than in OO and NN reports, but no difference was observed between OO and NN reports, $F(2, 44) = 4.18$, $p = .020$; multiple comparisons, $p_{LLvsOO} = .020$, $p_{NNvsLL} = .038$, $p_{NNvsOO} = .738$. OO reports also had higher scores for anticipation and aggression compared to NN reports, but LL scores did not differ from NN nor OO reports, anticipation, $F(2, 44) = 7.52$, $p = .0027$; multiple comparisons, $p_{NNvsOO} = .0018$, $p_{LLvsOO} = .142$, $p_{NNvsLL} = .339$; aggression, $F(2, 44) = 14.94$, $p < .0001$; multiple comparisons, $p_{NNvsOO} < .0001$, $p_{LLvsOO} = .141$, $p_{NNvsLL} = .106$. Regarding sadness, LL reports showed less score than OO reports, but no significant differences were found between the other type of reports, $F(2, 44) = 3.73$, $p = .029$; multiple comparisons, $p_{NNvsOO} = .482$, $p_{LLvsOO} = .025$, $p_{NNvsLL} = .119$. In terms of love, LL and OO reports scored higher than NN reports, but no differences were observed between LL and OO reports, $F(2, 44) = 7.67$, $p = .0005$; multiple comparisons, $p_{NNvsOO} = .029$, $p_{LLvsOO} = 1.000$, $p_{NNvsLL} = .008$. Regarding fun, there appears to be differences between the OO and LL groups, with OO reporting lower scores. However, no significant differences were observed among the other groups, $F(2, 44) = 3.73$, $p = .029$; multiple comparisons, $p_{NNvsOO} = .481$, $p_{LLvsOO} = .025$, $p_{NNvsLL} = .119$.

Considering that OBEs often involve distinct physical sensations, movements, and the nocturnal environment (Herrero et al., 2023), we undertook an analysis of

Figure 1
Analysis of Dream Content Using Emotion Lexicons



Note. (A) NRC Emotion Lexicon calculation of negative (left) and positive (right) emotions score for typical dream comparison. (B) Comparison of specific emotions and feelings score for typical dreams using the NRC Emotion Lexicon analysis. (C) Empath categories score for negative and positive emotions only for type of dreams. (D) Comparison of specific emotions and feelings score for typical dreams using the Empath category analysis. (E) Empath categories score of “body,” “movement,” and “night” for type of dreams only. (F) Comparison of specific emotions and feelings score for non-LDs using the NRC Emotion Lexicon analysis. (G) Comparison of specific emotions and feelings score for non-LDs using the Empath category analysis. (H) Total word count for typical dreams (left) and nonlucid dreams (right). The light gray asterisks refer to the nonlucid versus OBE comparison, while the dark asterisks refer to the lucid versus OBE comparison. NRC = National Research Council Canada; NN = non-LD reports of nonlucid dreamers; LL = LD reports of lucid dreamers; OO = OBE reports of OBE dreamers; OBE = out-of-body experience; Ang. = anger; Sad. = sadness; Antic. = anticipation; Disg. = disgust; Suffer = suffering; Agress = aggression; Surp. = surprise.

p values refer to Tukey-corrected multiple comparisons: * $p < .05$. ** $p < .01$. *** $p < .001$. **** $p < .0001$.

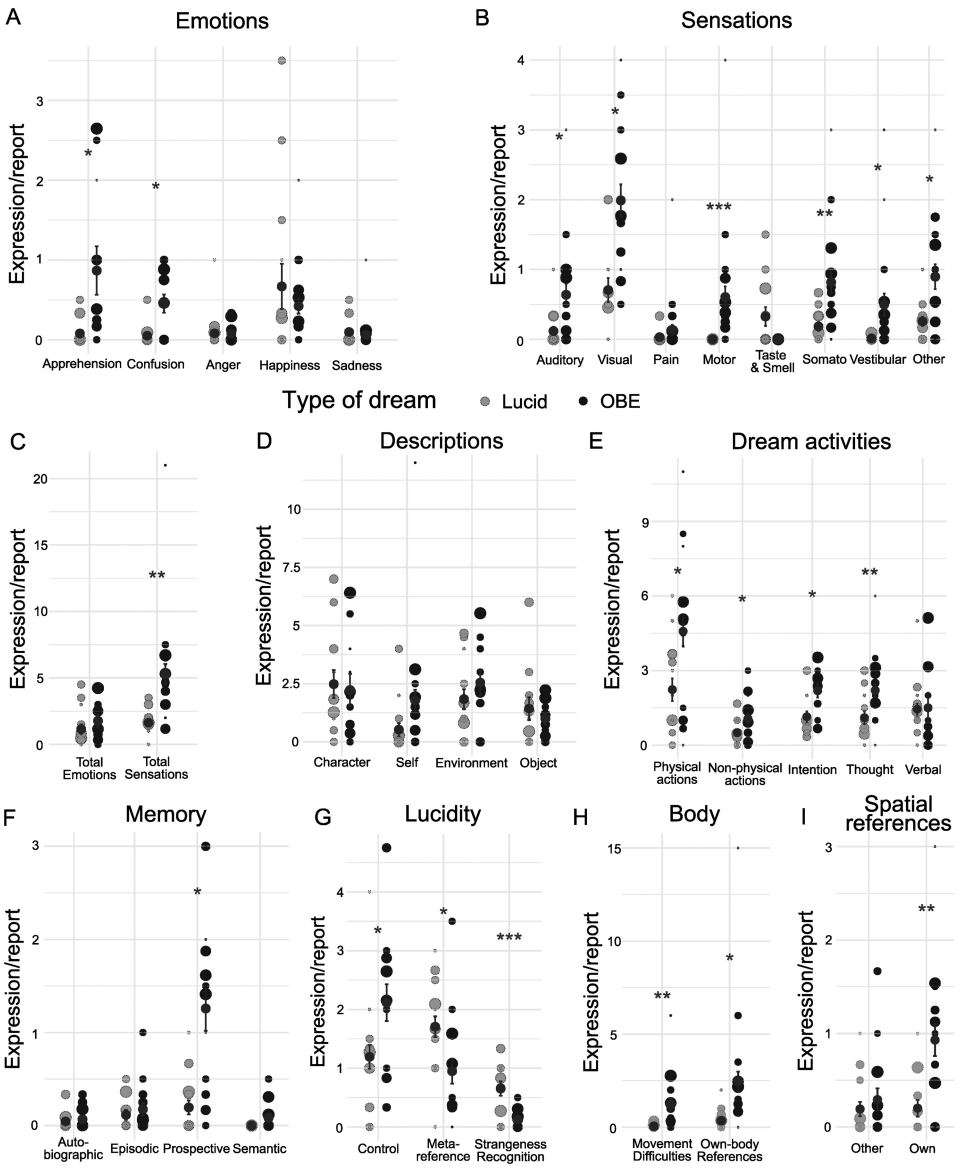
additional categories from Empath associated with pertinent thematic elements, specifically focusing on “movement,” “body,” and “night” (Figure 1E). The choice of these categories was driven by the inherent characteristics of OBEs and LDs. We found that OO reports had higher scores for expressions related to the body and movement than NN and LL reports, but no differences were found between NN and LL reports, body, $F(2, 44) = 18.32, p < .0001$; multiple comparisons, $p_{NNvsOO} < .0001, p_{LLvsOO} < .0001, p_{NNvsLL} = .658$; movement, $F(2, 44) = 13.65, p = .0001$; multiple comparisons, $p_{NNvsOO} < .001, p_{LLvsOO} < .001, p_{NNvsLL} = .943$, which is in line with the nature of OBEs (Blackmore, 1982; Herrero et al., 2023). Additionally, both LDs and OBEs had higher values for the category “night” compared to NN reports, but no significant differences were found between LL and OO reports, $F(2, 44) = 12.81, p < .0001$; multiple comparisons, $p_{NNvsOO} < .001, p_{NNvsLL} < .0001, p_{LLvsOO} = .051$, which could be related to the nature of the dream world and meta-references expected in lucid experiences (Baird et al., 2019; Mallett & Ashokkumar, 2022; Mallett et al., 2021; Stumbrys et al., 2014; Voss et al., 2013; Windt & Metzinger, 2007). It is important to highlight that the three conditions had similar total WC, NNs: 217.3 ± 53.2 words, LLs: 327.5 ± 97.38 words, OOs: 274.3 ± 92.1 words; $F(2, 44) = 1.63, p = .202$; multiple comparisons, $p_{NNvsOO} = .601, p_{LLvsOO} = .702, p_{NNvsLL} = .180$. Overall, OBE reports were more negatively emotionally charged than LDs and non-LDs, as they had higher scores for negative emotions and lower scores for positive emotions in a broad sense.

In order to control that differences found were due to the distinctive type of experiences and not to disparities in how each type of dreamer described the dream content, we analyzed the non-LDs reports of each type of dreamer. That is, the non-LD NN, NL, and NO. We found no significant difference in emotions calculated with the NRC emotion lexicon (Figure 1F and Table 1 in the online supplemental materials) or in the Empath analysis (Figure 1G and Table 2 in the online supplemental materials). In addition, there was no significant difference in the total WC (Figure 1H); NN: 217.3 ± 53.2 words, NL: 234.5 ± 44.4 words, NO: 169.7 ± 46.3 words, $F(2, 53) = 2.15, p = .120$.

Next, we created two categories related to dream awareness and dream control, named “insight” and “agency,” respectively, two cognitive notions that are widely accepted as typical components of LD (Baird et al., 2019; Mallett & Ashokkumar, 2022; Mallett et al., 2021; Stumbrys et al., 2014; Voss et al., 2013; Windt & Metzinger, 2007). Thus, to analyze dream reports based on these categories, we used the KH Coder 3 category-level analysis (similar to Mallett & Ashokkumar, 2022). For both categories, we found that more OO reports had related words for these categories than the NN and LL reports and more LL reports had related words than NN (words related to insight in, 39.90% of NNs, 54.90% of LLs, and 90.57% of OOs; Kruskal–Wallis statistic 43.54, $p < .0001$; corrected multiple comparison, $p_{NNvsOO} < .0001, p_{LLvsOO} = .0004, p_{NNvsLL} = .120$; words related to agency in, 57.14% of NNs, 66.67% of LLs, and 96.23% of OOs, Kruskal–Wallis statistic 28.34, $p < .0001$; corrected multiple comparison, $p_{NNvsOO} < .0001, p_{LLvsOO} = .001, p_{NNvsLL} = .201$).

We finally conducted a manual scoring using a simplified version of Hall and Van de Castle’s dream content scoring system (Domhoff, 1996). This analysis focused specifically on LL and OO reports (Figure 2). We observed that emotions such as confusion and apprehension were reported more often in OBEs than LDs (Figure 2A), $F(1, 27) = 5.39, p = .028$; $F(1, 27) = 5.19, p = .029$, respectively (Table 3 in the online

Figure 2
Results of the Manual Scoring of Dream Content Comparing Lucid Dreams and OBEs



Note. (A to I) Quantified expressions per report in different categories. (A) Emotions, (B) physical sensations and “others,” (C) total quantification of emotions and sensations, (D) descriptions, (E) dream activities, (F) type of memories, (G) lucidity elements, (H) body-related elements, and (I) and spatial references. OBE = out-of-body experience. p values are derived from the F test: * $p < .05$. ** $p < .01$. *** $p < .001$.

supplemental materials). Moreover, OBE reports had significantly more expressions per report related to visual, auditory, motor, somatosensory, vestibular, and “other” sensations (Figure 2B), $F_s(1, 27) > 5.19$, $P_s < .030$ (Table 3 in the online

supplemental materials). We found no differences in pain, $F(1, 27) = 1.42, p = .250$, nor in “taste and smell,” $F(1, 27) = 3.47, p = .076$. Interestingly, total sensations were higher in OBEs (Figure 2C), $F(1, 27) = 13.77, p = .002$ than in LDs. We did not find significant differences in descriptions of any kind, either of themselves, characters, environment, or objects (Figure 2D), $F(1, 27) > 1.79, p > .190$ (Table 3 in the online supplemental materials). Of all the dream activities that we quantified, we recorded a higher number of expressions in OBEs compared to LDs for thoughts, physical activities, nonphysical activities, and intentions (Figure 2E), $F(1, 27) > 5.42, p < .028$ (Table 3 in the online supplemental materials), but no differences were found for verbalizations as a dream activity (Table 3 in the online supplemental materials). Regarding mnemonic references (Figure 2F), we quantified the expressions that referred to some retrieval that occurred during the experience. OBEs reports showed greater amounts of prospective memory expressions, $F(1, 27) = 4.62, p = .040$, than LDs, and there were no differences in other types of memory, $F(1, 27) > 0.10, p > .755$ (Table 3 in the online supplemental materials). In terms of lucidity (Figure 2G), LDs had more meta-references, references to the dreamer’s awareness of being in a dream, $F(1, 27) = 4.98, p = .038$, and recognition of strange occurrences during the dream than OBEs, $F(1, 27) = 18.87, p = .0002$. In contrast, OBEs were more likely to involve effective control of the dream, $F(1, 27) = 7.35, p = .026$, presented more references to the dreamer’s own body, $F(1, 27) = 7.72, p = .013$, and more references to difficulties in movement within the dream environment (Figure 2H), $F(1, 27) = 8.03, p = .007$, indicating challenges in navigating or moving around within the dream space. Additionally, there were significant differences in the spatial references addressed to oneself, indicating that individuals tended to refer to themselves using more expressions such as “to my left” or “behind me” than in LDs (Figure 2I), $F(1, 27) = 9.03, p = .005$, but no significant differences were found in the spatial references addressed to other characters in the dream, $F(1, 27) = 0.46, p = .505$.

Discussion

Here, we showed that OBE reports can be distinguished from reports of both lucid and non-LD in several ways. These include variations in emotional scoring, the sense of agency and insight, and the categories of dream elements such as “body,” “movement,” and “night.” Additionally, differences were observed in the sensory experiences, dream activities, memory references, aspects of lucidity, and the body and spatial references utilized in the reports.

First, OBEs were characterized by a higher incidence of words related to negative emotions, such as fear, disgust, suffering, anticipation, and aggression (Figure 1). This suggests that OBEs tend to have a greater prevalence of negative emotions. The observed increase in negative emotions could be related to the fact that OBEs occur during SP. Recently, Herrero et al. (2023) compared the perceived emotions during the SP episodes with and without OBE episodes. SP episodes without OBE had more negative and less positive emotions than OBE episodes. Similar results were published by Kliková et al. (2021) who showed that vestibular–motor hallucinations during SP generate more pleasant experiences than other SP hallucinations such as the intruder or incubus.

Second, more OBE reports contained words related to agency and insight than lucid and non-LD reports. Agency and insight are crucial elements of lucid dreaming

as they indicate dream control and awareness, respectively. These concepts are commonly found in the literature on lucid dreaming and are considered necessary components of such dreams (Baird et al., 2019; Mallett & Ashokkumar, 2022; Mallett et al., 2021; Stumbrys et al., 2014; Voss et al., 2013; Windt & Metzinger, 2007). This result suggests that OBE experiences may involve even greater control and self-awareness than LD. However, the terms in the “agency” and “insight” categories may not explicitly indicate “dream control” or “dream awareness.” For example, the term “sentir” (to feel) in the “insight” category has varied interpretations beyond direct awareness, affecting result interpretation. This is particularly relevant as OBE dreamers reported more sensations (excluding “pain” and “taste and smell,” Figure 2B and 2C). The potential bias is not intrinsic to insight but arises from the analysis in Figure 2, revealing more “sensations” in OBE reports, inherently associated with the verb “feel.” OBEs also reported more dream activities (Figure 2E) and more references to prospective memories (Figure 2F), which is in line with higher dream control, as well as more spatial (Figure 2I) and body references and difficulties with movement in the dream environment (Figure 2H). OBEs can manifest in different states of consciousness, such as wakefulness (Ehrsson, 2007; Guterstam & Ehrsson, 2012; Slater et al., 2010; Wu et al., 2023), sleep (Blackmore, 1982; de Sá & Mota-Rolim, 2016; Irwin, 1988; S. P. LaBerge, 1988; S. A. Mota-Rolim et al., 2020). It is reasonable to suggest that if OBEs during sleep involve the activation of the same cortical regions as those during wakefulness, the temporoparietal cortex may also play a role in sleep-related OBEs. This could account for the greater number of sensations and level of dream control reported by OBE dreamers, indicating a heightened level of awareness and engagement with the dream environment. The temporoparietal junction is a versatile brain region that combines various sensory information and contributes to self-awareness and internal body imaging (Blanke & Mohr, 2005). Therefore, according to our results, OBEs can be viewed as a more actively experienced form of dreaming than non-LD and LD. OBEs are characterized by an increased frequency of bodily and spatial references, as well as difficulties with movement, which emphasize physical sensations and spatial orientation, as our analysis reveals.

Here, we found that OBEs and LDs share some similarities, such as a heightened level of awareness and greater control over the dream environment. However, the unique sensations, references, and difficulties with movement found in OBEs strongly suggest that these experiences are not simply a variation of LD.

Furthermore, it is worth noting that educational levels of participants were not specifically controlled for, which could potentially influence their ability to express thoughts and experiences using richer vocabulary. However, despite this limitation, we also analyzed non-LD from a variety of dreamers and found no differences between them, which suggest that the differences observed between different types of dreams are not due to individual factors, as the same individuals had both the nonlucid and OBE dreams analyzed. We rigorously controlled quantitative linguistic measures and found no evident differences among the text corpora (refer to Table 1). These findings suggest that OBEs and LDs may involve different cognitive processes and may be experienced differently. Further research is needed to fully understand the distinctions between these two phenomena and their implications for studying human consciousness and sleep-dependent processes.

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